

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Report Analysis Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. **(20 Marks)**

Commands :

1. nmap scanme.nmap.org
2. nmap -sV scanme.nmap.org
3. sudo nmap -O scanme.nmap.org
4. sudo nmap -A scanme.nmap.org

Observation :

- The host is alive.
- HTTP (80) is open.
- SSH (22) is open.
- SIP ports 5060 and 5061 are open.
- Operating system information identified.

Security Recommendations :

- Keep all software updated.
- Enable firewall protection.

- Use strong authentication mechanisms.
- Perform periodic vulnerability assessments.
- Conduct regular penetration testing.



Report : Question1_Nmap_Report.txt

```

kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
--(kali@kali)~--
$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

--(kali@kali)~--
$ nmap scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:40 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 88.31 seconds

--(kali@kali)~--
$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:43 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.004s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
53/tcp    closed domain
80/tcp    open  http        Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8085/tcp  closed unknown

```

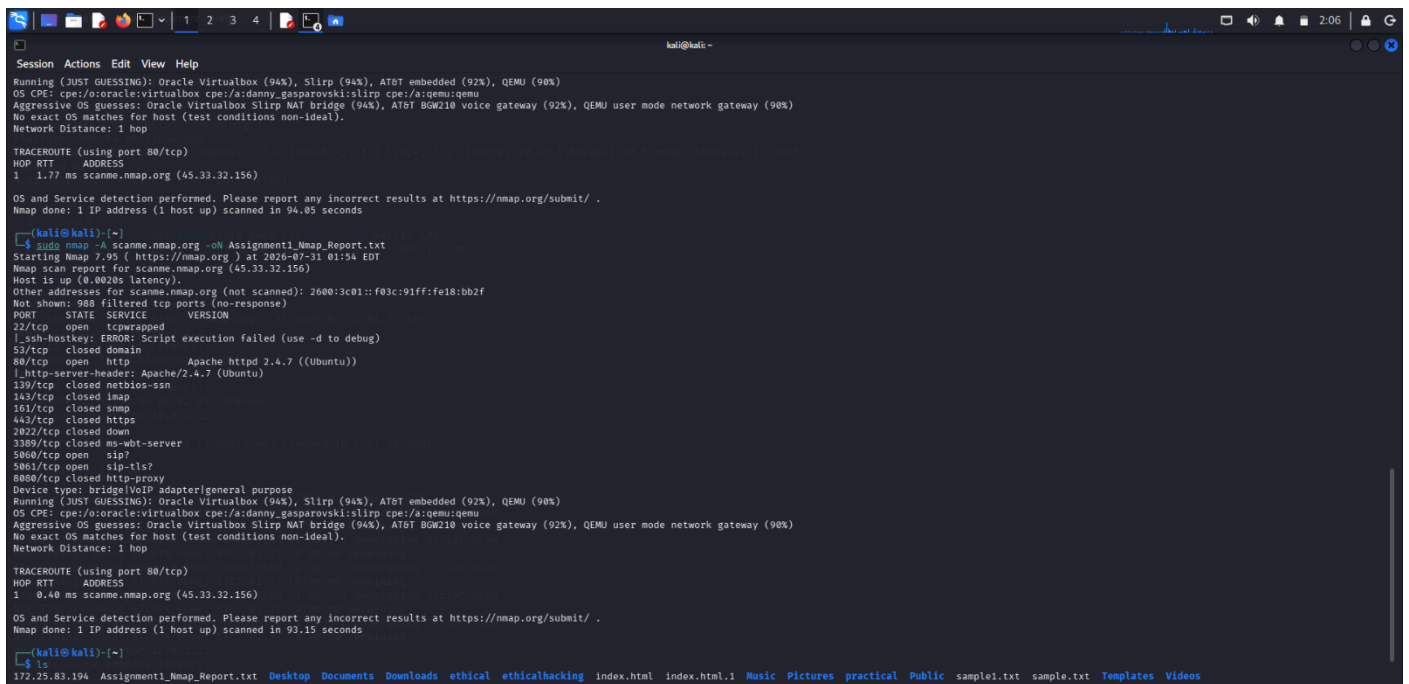
```

kali@kali: ~
Session Actions Edit View Help
$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:49 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
83/tcp    closed mt-ml-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
5000/tcp  closed isakmp
2022/tcp  closed down
3120/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds

--(kali@kali)~--
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:51 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), SLiRP (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovskis:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox SLiRP NAT bridge (94%), AT&T Bgw210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

```



```
kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0020s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 888 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  tcpwrapped
1_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http
1_http-server-header: Apache/2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2802/tcp  closed down
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.77 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.05 seconds

kali@kali:~$ ls
772.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Procedure :

1. Created Email Template.
2. Created Landing Page.
3. Configured Mailtrap SMTP.
4. Created User Group.
5. Created Campaign.
6. Send phishing emails.
7. Viewed campaign statistics.

Email Security Controls :

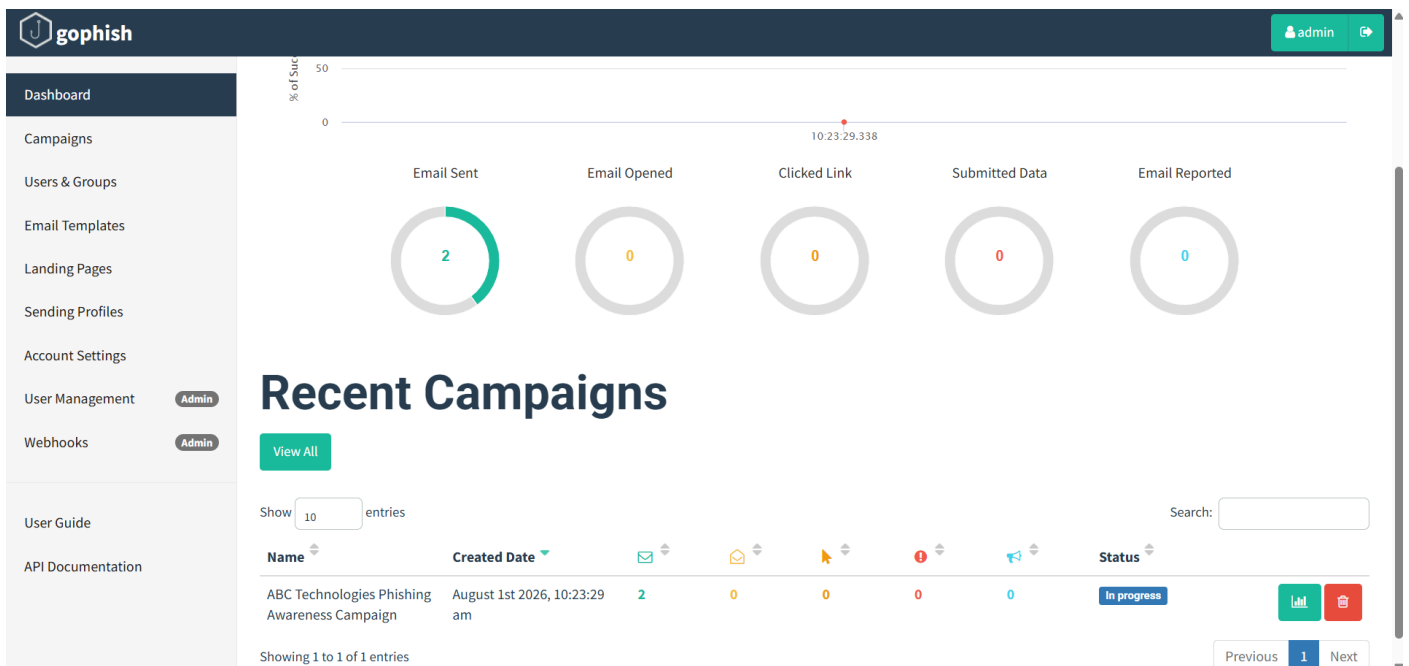
- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).

- Configure DKIM (DomainKeys Identified Mail).
- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).
- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering.



ABC Technologies
Phishing Awareness C

Report :



Edit Group

Name: Tharun Venkat

[+ Bulk Import Users](#) [Download CSV Template](#)

First Name: Last Name: Email: Position: [+ Add](#)

Show 10 entries Search:

First Name	Last Name	Email	Position
Bhogyam	Venkat	bhogyamtharun...	
Nithin	Varma	tharun.bv5@gm...	

Showing 1 to 2 of 2 entries Previous 1 Next

[Close](#) [Save changes](#)

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure..(10 Marks)

Command :

dig AXFR zonetransfer.me @nsztml.digi.ninja

Observation :

- DNS Zone Transfer succeeded.
- Internal DNS records were exposed.
- Mail server records were visible.
- Multiple hostnames and IP addresses were disclosed.

Possible Attacks :

An attacker can use the exposed information to:

- Perform reconnaissance.
- Scan exposed servers.
- Launch phishing attacks.
- Target mail servers.
- Exploit vulnerable web servers.
- Attempt brute-force attacks on exposed services.
- Prepare for advanced cyberattacks.

To Secure DNS Infrastructure :

- Disable DNS Zone Transfer for unauthorized users.
- Allow AXFR only between trusted DNS servers.
- Configure Access Control Lists (ACLs).
- Enable DNSSEC to protect DNS integrity.
- Regularly update DNS software.
- Monitor DNS logs.

- [illegible]

Report :



Assignment3_DNS_AX
FR_Report.txt

Code of Conduct:

I K.Nithin varma (192312272) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Nithin varma

Signature of the student:

RUBRICS FOR CALCULATION

Report Analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Report Analysis and Interpretation	25%	Accurately interprets the security report, identifies all key findings, and explains their significance with excellent technical understanding.	Correctly interprets most findings with minor omissions or inaccuracies.	Identifies only some findings and provides limited interpretation.	Unable to interpret the report or identify key findings.
Identification of Security Risks and Vulnerabilities	25%	Correctly identifies all security risks, vulnerabilities, and possible attack vectors with clear technical justification.	Identifies most risks and vulnerabilities with minor errors.	Identifies only a few risks with limited technical explanation.	Fails to identify major risks or vulnerabilities.
Application of Cybersecurity Concepts and Countermeasures	30%	Applies appropriate ethical hacking concepts and recommends comprehensive, practical, and technically sound countermeasures.	Recommends suitable countermeasures with minor technical gaps.	Provides limited or partially appropriate recommendations.	Recommendations are inappropriate, incomplete, or missing.
Technical Communication and Documentation	20%	Response is well-organized, technically accurate, clearly presented, and uses appropriate cybersecurity terminology throughout.	Response is organized and understandable with minor documentation or presentation issues.	Response lacks clarity, organization, or contains several technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.